

Cuong Bui

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Berkeley-trained physicist with experience in machine learning, scientific computing, and software development. Passionate about applying computational methods to solve engineering and scientific challenges in mission-driven technologies.

Education

UC Berkeley

December 2024

B.A. in Astrophysics and High Honors in Physics. GPA: 3.86/4.00

Relevant Coursework: Multivariable Calculus, Linear Algebra and Differential Equations, Computer Simulations, Analytic Mechanics, Data Structures and Algorithms, Bayesian Data Analysis and Machine Learning, Concepts of Probability.

Work Experience

Software Product Developer

January 2026 - April 2026

Omnium LLC

- Developed a GitHub-based application launchpad that replaced existing internal tooling workflows and centralized deployment, version control, and distribution of internal analytical tools.
- Engineered a desktop application integrating waterfall chart analytics into an intuitive interface, streamlining internal analysis workflows.

Data Scientist Intern

May 2025 - December 2025

Omnium LLC

- Developed Python- and Excel-based analytical models to evaluate pricing scenarios, forecast demand, and identify competitive market trends using large-scale consumer data.
- Designed a machine learning framework for detecting outliers in time-series pricing models using DBSCAN and Isolation Forest algorithms.
- Integrated the outlier detection framework into a production application, enabling company-wide access to automated anomaly detection.

Student Researcher

January 2025 - May 2025

Physics Department, UC Berkeley

- Developed and trained a convolutional neural network (CNN) for classification of particle-detector noise time-series data.
- Presented deep learning research findings at collaboration meetings and American Physical Society (APS) conferences.

Undergraduate Student Instructor

August 2024 - December 2024

Astronomy Department, UC Berkeley

- Designed and taught weekly discussion sections, quizzes, and office hours for an introductory astronomy course serving over 700 students.

Department Student Advisor

January 2024 - December 2024

Astronomy Department, UC Berkeley

- Mentored undergraduate astronomy students on academic planning, research opportunities, and course selection in collaboration with faculty advisors.

Projects

Undergraduate Honors Thesis : "Using CNNs for Event Pileup Discrimination in CUPID"

August 2023 - December 2024

High Honors in Physics

"Using Convolutional Neural Networks for Event Pileup Discrimination in CUPID"

- Generated and analyzed over 100,000 simulated particle-detector signals to train and evaluate CNN models.
- Evaluated CNN architectures for event pileup discrimination, improving detector background noise rejection in the CUPID experiment.

Interests and Extracurriculars

Colorguard Performer

2019 - 2026

- Pacific Crest Drum and Bugle Corps (2019, 2021), Bluecoats Drum and Bugle Corps (2022, 2024), and Vox Artium Winter Guard (2026) performer. Performed with nationally ranked ensembles for collective audiences exceeding 300,000.

Skills

Programming: Python, Java, R

Machine Learning: TensorFlow, Scikit-learn, CNNs, Regression, Time-Series Analysis, DBSCAN, Isolation Forest

Software: Git, GitHub, Jupyter, VS Code, Excel, LaTeX, NumPy, Pandas, Matplotlib

Analysis: Statistical Modeling, Data Visualization, Signal Processing